

Isha Pali

Research Interests: Network Security, Automotive security, Adversarial machine learning

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Amlidih, Raipur, 492001 (C.G)

Education

Ph.D. in Computer Science and Engineering: University of Queensland – IIT Delhi Research Academy (**UQIDRA**)

Supervisors: [Dr. Dan Kim \(UQ\)](#), [Dr. Vireshwar Kumar \(IITD\)](#), [Dr. Naipeng Dong \(UQ\)](#)

M.Tech in Computer Science Engineering: DSPM IIIT-Naya Raipur | CGPA: 9.4 / 10 | Silver Medalist (Highest CGPA)

Supervisor: [Dr. Ruhul Amin \(IIITNR\)](#)

B.Tech in Computer Science Engineering NIIT University, Rajasthan | CGPA: 9.47 / 10

Minor in Finance (CGPA: 9.67 / 10)

Technical Skills

Core Technologies & AI/ML: Python, C++, P4, PyTorch, scikit-learn, Adversarial Robustness Toolbox (ART)

Systems & Networking: SDN, BMv2, Mininet, Wireshark, CARLA, ProVerif, Scyther

Awards

- Won the **Best Poster award and cash prize of INR 3k** at UQ-IITD Annual Symposium 2026.
 - **GATE qualified** — Score: 418, Rank: 8022
 - **participated in the Grand Finale** — Smart India Hackathon 2018 held in Kolkata.
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Industry & Teaching Experience

TATA motors project | funded automotive security research | IIT Delhi Jan 2025 - present

- Designed and evaluated adversarial attacks against ML-based CAN bus IDS, validated in the CARLA autonomous driving simulator.
- Developing defence mechanisms to improve the robustness of in-vehicle intrusion detection against adversarial attacks.

Academic Tutor | The University of Queensland Jan 2024 – Dec 2024

- Tutored two courses across two semesters for undergraduate and postgraduate students at UQ
 - *CYBR3000 Information Security*
 - *COMS3200 Computer Networks*

Intern (Cyber Security Analyst) | PricewaterhouseCoopers (PwC) — Jan 2019 – July 2019

- Built a Python/Selenium automation pipeline that replaced a manual report generation and delivery workflow, reducing turnaround time significantly.
 - Developed employee security awareness training material aligned to the client's Code of Ethics and Information Security Policies.
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Publications

Journals

- **Pali, I.**, Abdussami, M., Amin, R., Al-Shehari, T., Al-Razgan, M., & Alfakih, T. (2024). S2DN: Design of robust authentication protocol with session key establishment in multi-controller based software-defined VANETs. ***Vehicular Communications***, 47, 100767.
- **Pali, I.**, Amin, R., & Abdussami, M. (2024). Autonomous vehicle security: Current survey and future research challenges. ***Security and Privacy***, 7(3), e367.
- Amin, R., **Pali, I.**, & Sureshkumar, V. (2021). Software-defined network enabled vehicle to vehicle secured data transmission protocol in VANETs. ***Journal of Information Security and Applications***, 58, 102729.

Conference

- **Pali, I.**, Thakur, S., Maurya, V., Dong, N., Kumar, V., & Kim, D. D. Poster: Mind the Structural Gap-Evaluating Automotive CAN Datasets as Interchangeable Benchmarks for Intrusion Detection, **Usenix Vehiclesec'26**
- **Pali, I.**, & Amin, R. (2022, May). PortSec: Securing port knocking system using sequence mechanism in SDN environment. In 2022 ***International Wireless Communications and Mobile Computing (IWCMC)*** (pp. 1009-1014). IEEE. (core B)

Preprint / Under review

- **Pali, I.**, Krishania, L., Chadha, D., Kandar, A., Varshney, G., & Shukla, S. (2020). A Comprehensive Survey of Aadhar and Security Issues. *arXiv:2007.09409*.
- PLAID: Practical Machine Learning-Assisted Robustness Framework against Automotive Intrusion Detection. Submitted to **ASIA CCS 2027** (CORE A), currently under review.

Talks

- **Guest for an Invited talk** at **Oracle Labs Brisbane**, Australia on **14 December 2024**.
- Presented a talk at **UQ Cyber Security Research Conference (CSRC 2024)**, **11 July 2024**.
- Presented my research at **UQ – IITD Knowledge Nexus Seminar Series** - India-Australia Energy Corridor.

Research Summary

PhD candidate at the University of Queensland-IIT Delhi Research Academy specializing in adversarial machine learning, AI security, and in-vehicular network security. Experienced in designing adversarial robustness frameworks and evaluating intrusion detection systems for autonomous driving environments. Strong background in research, teaching, and industry consulting through roles at PwC and academia. Interested in applying secure and trustworthy AI research to real-world intelligent systems.

Research Experience

Ph.D. Researcher | 2018 – present

Adversarial attacks and defenses in in-vehicular networks (ongoing)

- Designed an adversarial robustness framework targeting CAN bus intrusion detection systems, generating inputs that bypass detection while preserving physical plausibility.
- Implemented and validated the proposed strategy in the CARLA autonomous driving simulator, testing across multiple IDS architectures.
- Evaluating vulnerabilities in automotive IDS architectures to design defense frameworks, with ongoing work aimed at securing edge-based AI deployments.

PortSec: Port Knocking Authentication in SDN

- Proposed a novel communication protocol using port knocking as an authentication mechanism in SDN-enabled networks, with informal security analysis.

- Implemented and tested the protocol in P4 on a BMv2 virtual switch.

Research Intern | June 2018 – July 2018 DSPM IIIT Naya Raipur — under Dr. Ruhul Amin

- Studied security challenges in SDN-based Vehicular Ad-hoc Networks (VANETs).
- Proposed a secure communication protocol for SDN-enabled V2V data transmission; work led to a journal publication in JISA (IF 4.96).

Statement of Motivation

Dear Selection Committee,

I am excited to apply for the Makers Lab Internship Program at Tech Mahindra. As a **PhD** candidate in the **University of Queensland–IIT Delhi Research Academy**, my research focuses on adversarial machine learning and the security of in-vehicular networks. Through this work, I have developed expertise in identifying vulnerabilities in AI-driven systems and designing robust security frameworks for autonomous environments.

I am particularly drawn to Makers Lab because of its commitment to translating cutting-edge research into impactful real-world solutions, as demonstrated by initiatives such as Entellio and its work on open-source AI models. The Lab's focus on AI, Edge AI, and intelligent systems aligns closely with my research interests and long-term goal of developing secure, trustworthy, and resilient AI technologies.

In addition to my research experience, my roles as an Academic Tutor and as an Analyst Intern at PwC have strengthened my ability to communicate complex technical concepts and collaborate effectively across diverse teams. I am eager to contribute my expertise in adversarial machine learning and AI security to Makers Lab's research initiatives while gaining experience in translating advanced research into scalable, real-world AI solutions.

Sincerely,

Isha Pali
PhD at UQ-IITD Research Academy